Government Engineering College Sheikhpura

B. Tech First Semester Mid-Term Examinations 2025

Branch- Civil Engineering | Full Marks-20

Subject/Code- Engineering Mechanics/100105 | Time- 2 Hrs.

Instructions

(i) **There are 7 questions in section A and 7 questions in section B.**

(ii) **Attempt 5 questions from each section.**

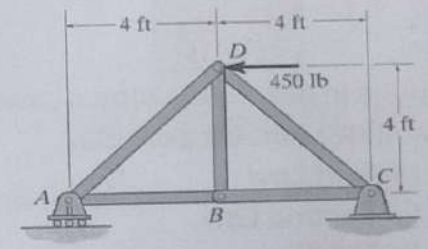
(iii) **Distribution of marks for each question is given at right hand margin.**

(iv) **Any missing data and assumptions can be made suitably.**

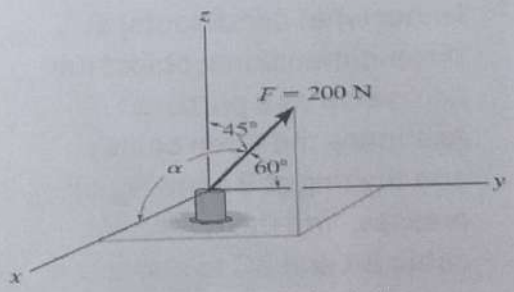
SECTION-A

1. Differentiate between free vector and sliding vector with respect to their lines of action and points of application. 1

2. Explain either theoretically or numerically,why the **member BD** of the truss shown below is a **Zero-force member.** 1



3. Find the value of angle α in degree for the force vector F shown in the below figure. 1



4. The centre of gravity is based on the application of principle of_____(the extension of varignon's theorem in three dimension). 1

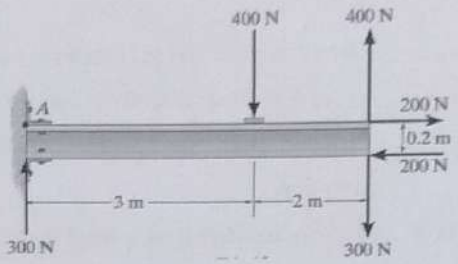
5. Write **True** or **False** for the following statement. 1
Statement: Simple trusses follow the relation m=2×n-3
Where, m=no. of members and n=no. of joints.

6. Identify a vector quantity from the quantities listed below 1

(a) Density
(b) Electric Current
(c) Finite angular displacement
(d) Infinitesimally small angular displacement

7. 1

Determine the resultant couple moment acting on the beam.

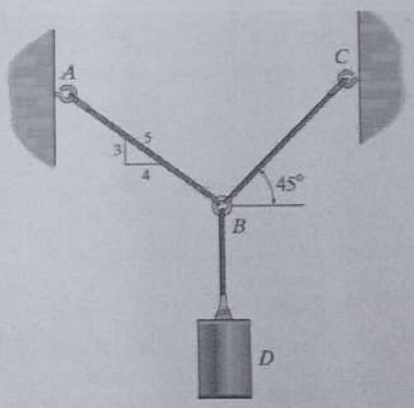


SECTION-B

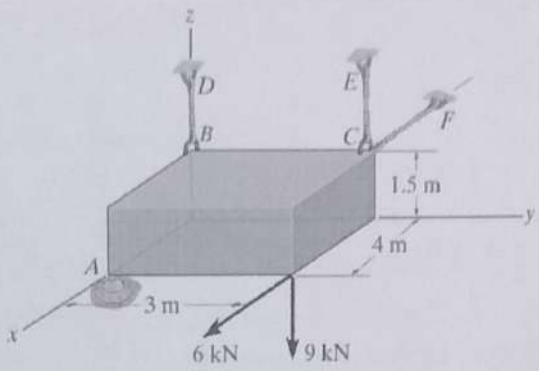
1. Explain in **details** Newton's Laws of Motion with the help of example wherever possible 1+1+2

(a) First Law
(b) Second Law
(c) Third Law

2. Under what condition(s) a three-dimensional object can be treated as a particle. Assuming the three cables and the ring at B of negligible masses, find the tension in cable BA and BC to support a cylinder of 200kg. 0.5+2.5

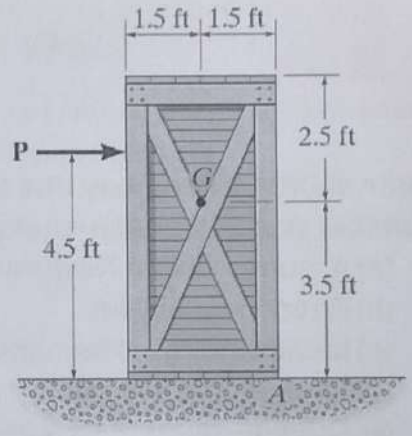


3. Determine the force developed in the cords BD,CE, and CF and the reactions of the block on the ball and socket joint at A. 3

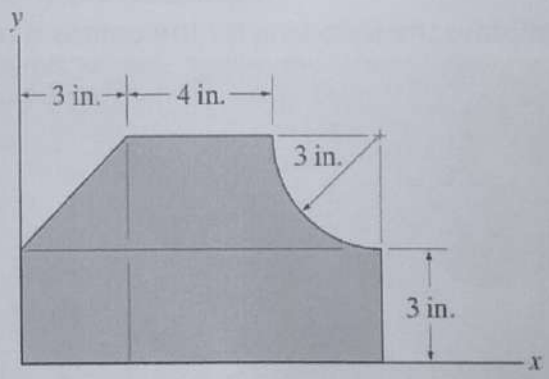


4. 3

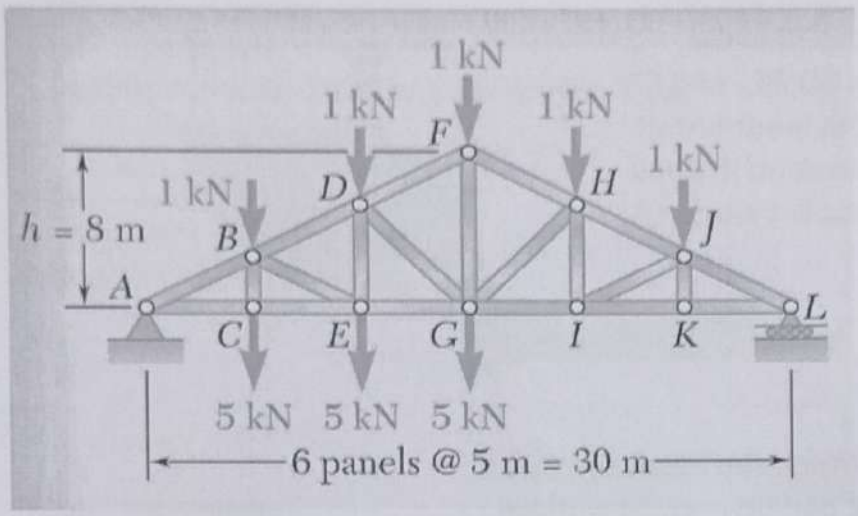
Determine the minimum force P that can be applied without causing movement of the 250-lb crate which has a centre of gravity at G. the coefficient of static friction at the floor is μ_s=0.4.



5. Determine the location of the centroid of the composite area shown below. 3



6. Using the method of sections, find the force in members HJ, HI, and GI of the truss shown. 3

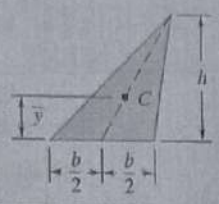
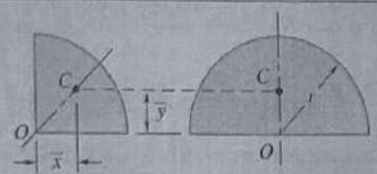


7. Write short notes on **any three** of the following with the help of a suitable sketch for each wherever possible. 3

(a) Assumptions in Newtonian mechanics
(b) Wrench Resultant
(c) Idealizations in Mechanics
(d) Three-Force members
(e) Couple-moment as a free vector.

Important Formulas

1. Consider the following for the centroid of an area.

Shape		$\bar{x}$	$\bar{y}$	Area
Triangular area			$\frac{h}{3}$	$\frac{bh}{2}$
Quarter-circular area		$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{4}$
Semicircular area		0	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{2}$